LAPORAN PRATIKUM TEKNOLOGI IOT 2

Tugas dan Latihan Module 5

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Agus ambarwari, S.Pd., M.Kom.

JURUSAN TEKNOLOGI INFORMASI

PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

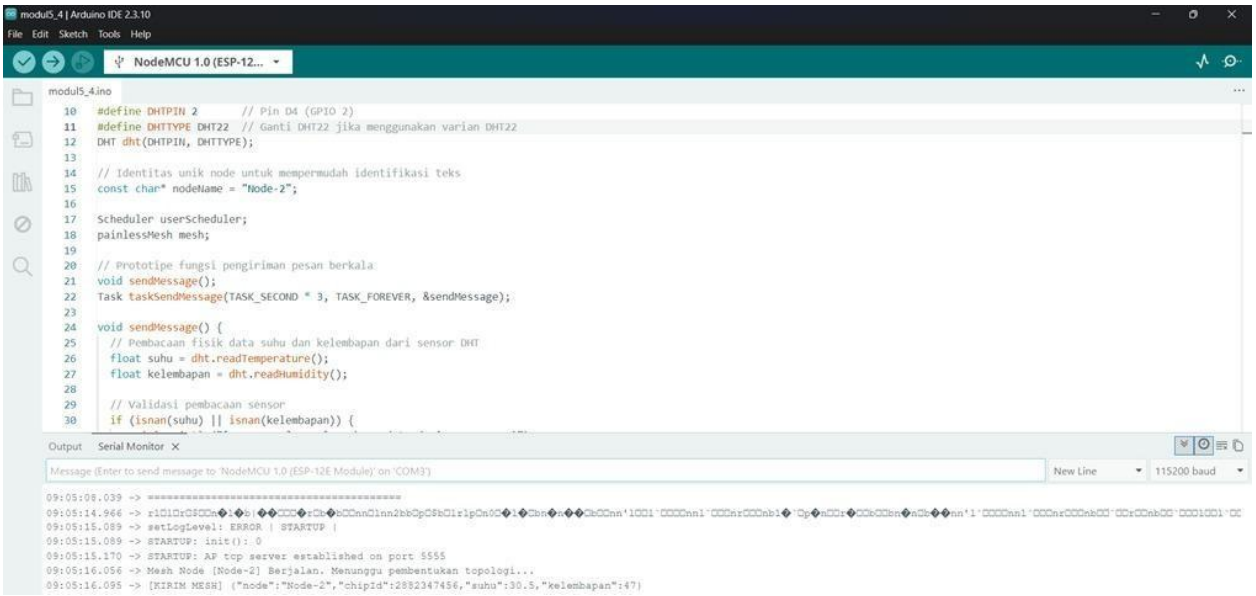
POLITEKNIK NEGERI LAMPUNG TAHUN

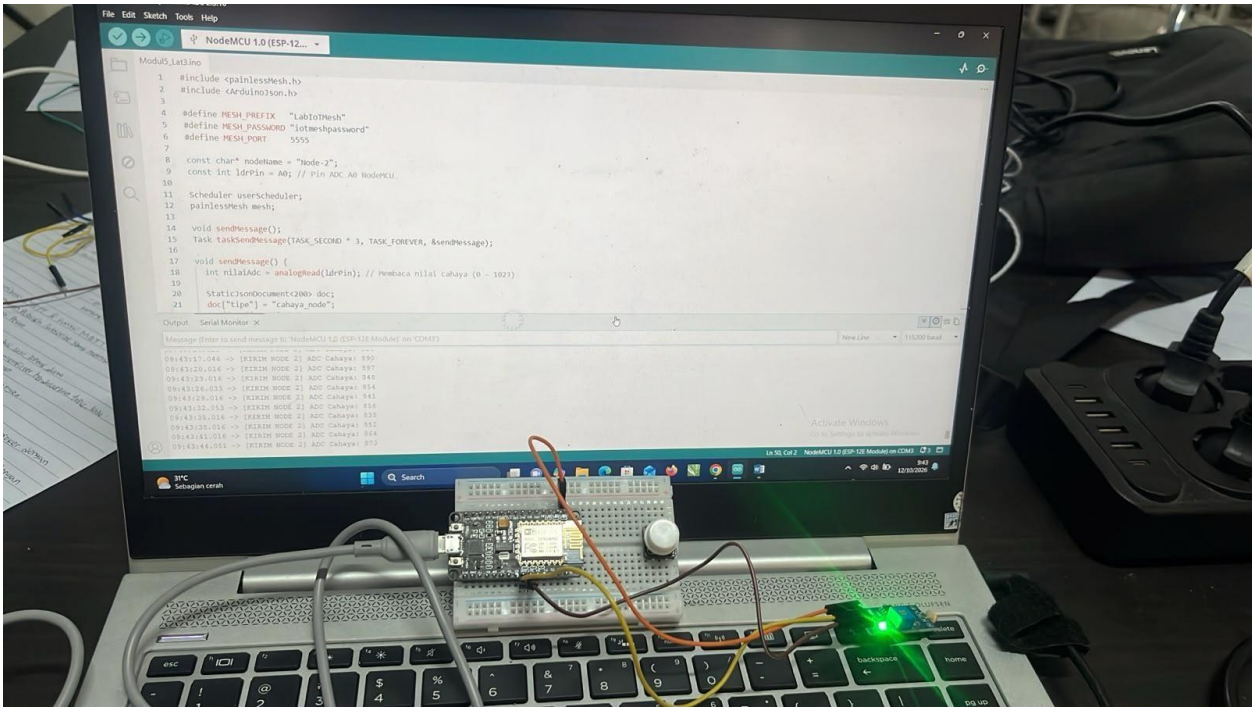
2025/2026

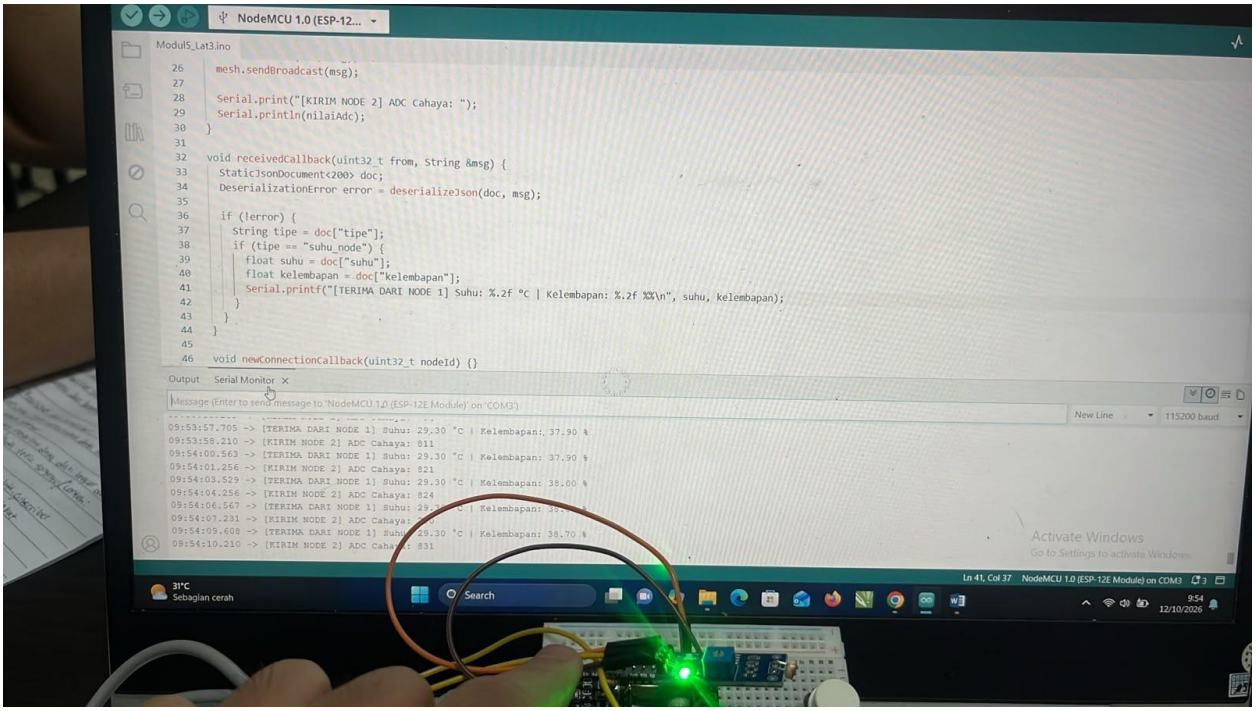
5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh

Kembangkan jaringan mesh pada Praktikum 4 menjadi sistem *monitoring* lingkungan dan kendali otomasi terdistribusi tanpa perantara router. Setiap kelompok membagi peran perangkat sebagai berikut:

Bukti :




File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
Modul5_Lat3.ino
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "LabIoTMesh"
#define MESH_PASSWORD "iotmeshpassword"
#define MESH_PORT 5555
const char* nodeName = "Node-2";
const int ldrPin = A0; // Pin ADC A0 NodeMCU
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
int nilaiAdc = analogRead(ldrPin); // Membaca nilai cahaya (0 - 1023)
StaticJsonDocument<200> doc;
doc["tipe"] = "cahaya_node";
Output Serial Monitor
09:43:17.046 -> [KIRIM NODE 2] ADC Cahaya: 890
09:43:20.016 -> [KIRIM NODE 2] ADC Cahaya: 897
09:43:23.016 -> [KIRIM NODE 2] ADC Cahaya: 845
09:43:26.033 -> [KIRIM NODE 2] ADC Cahaya: 854
09:43:29.014 -> [KIRIM NODE 2] ADC Cahaya: 841
09:43:32.053 -> [KIRIM NODE 2] ADC Cahaya: 856
09:43:35.016 -> [KIRIM NODE 2] ADC Cahaya: 835
09:43:38.016 -> [KIRIM NODE 2] ADC Cahaya: 852
09:43:41.016 -> [KIRIM NODE 2] ADC Cahaya: 864
09:43:44.051 -> [KIRIM NODE 2] ADC Cahaya: 873
Activate Windows
31°C
Sebagian cerah
Search
12/10/2026


NodeMCU 1.0 (ESP-12...
Modul5_Lat3.ino
mesh.sendBroadcast(msg);
Serial.print("[KIRIM NODE 2] ADC Cahaya: ");
Serial.println(nilaiAdc);
}
void receivedCallback(uint32_t from, String &msg) {
StaticJsonDocument<200> doc;
DeserializationError error = deserializeJson(doc, msg);
if (!error) {
String tipe = doc["tipe"];
if (tipe == "suhu_node") {
float suhu = doc["suhu"];
float kelembapan = doc["kelembapan"];
Serial.printf("[TERIMA DARI NODE 1] Suhu: %.2f °C | Kelembapan: %.2f %%\n", suhu, kelembapan);
}
}
}
void newConnectionCallback(uint32_t nodeId) {}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
09:53:57.705 -> [TERIMA DARI NODE 1] Suhu: 29.30 °C | Kelembapan: 37.90 %
09:53:58.210 -> [KIRIM NODE 2] ADC Cahaya: 811
09:54:00.563 -> [TERIMA DARI NODE 1] Suhu: 29.30 °C | Kelembapan: 37.90 %
09:54:01.256 -> [KIRIM NODE 2] ADC Cahaya: 821
09:54:03.529 -> [TERIMA DARI NODE 1] Suhu: 29.30 °C | Kelembapan: 38.00 %
09:54:04.256 -> [KIRIM NODE 2] ADC Cahaya: 824
09:54:06.567 -> [TERIMA DARI NODE 1] Suhu:
09:54:07.231 -> [KIRIM NODE 2] ADC Cahaya:
09:54:09.608 -> [TERIMA DARI NODE 1] Suhu: 29.30 °C | Kelembapan: 38.70 %
09:54:10.210 -> [KIRIM NODE 2] ADC Cahaya: 831
Activate Windows
Go to Settings to activate Windows.
Ln 41, Col 37 NodeMCU 1.0 (ESP-12E Module) on COM3
31°C
Sebagian cerah
Search
9:54
12/10/2026

tugas.ino

```ino
#include <painlessMesh.h>
#include <DHT.h>
#include <ArduinoJson.h>

#define MESH_PREFIX   "LabIoTMesh"
#define MESH_PASSWORD "iotmeshpassword"
#define MESH_PORT     5555

#define DHTPIN 2       // Pin D4 (GPIO 2)
#define DHTTYPE DHT22  // Sesuaikan jika menggunakan DHT11
DHT dht(DHTPIN, DHTTYPE);

const char* nodeName = "Node-1-haikal";

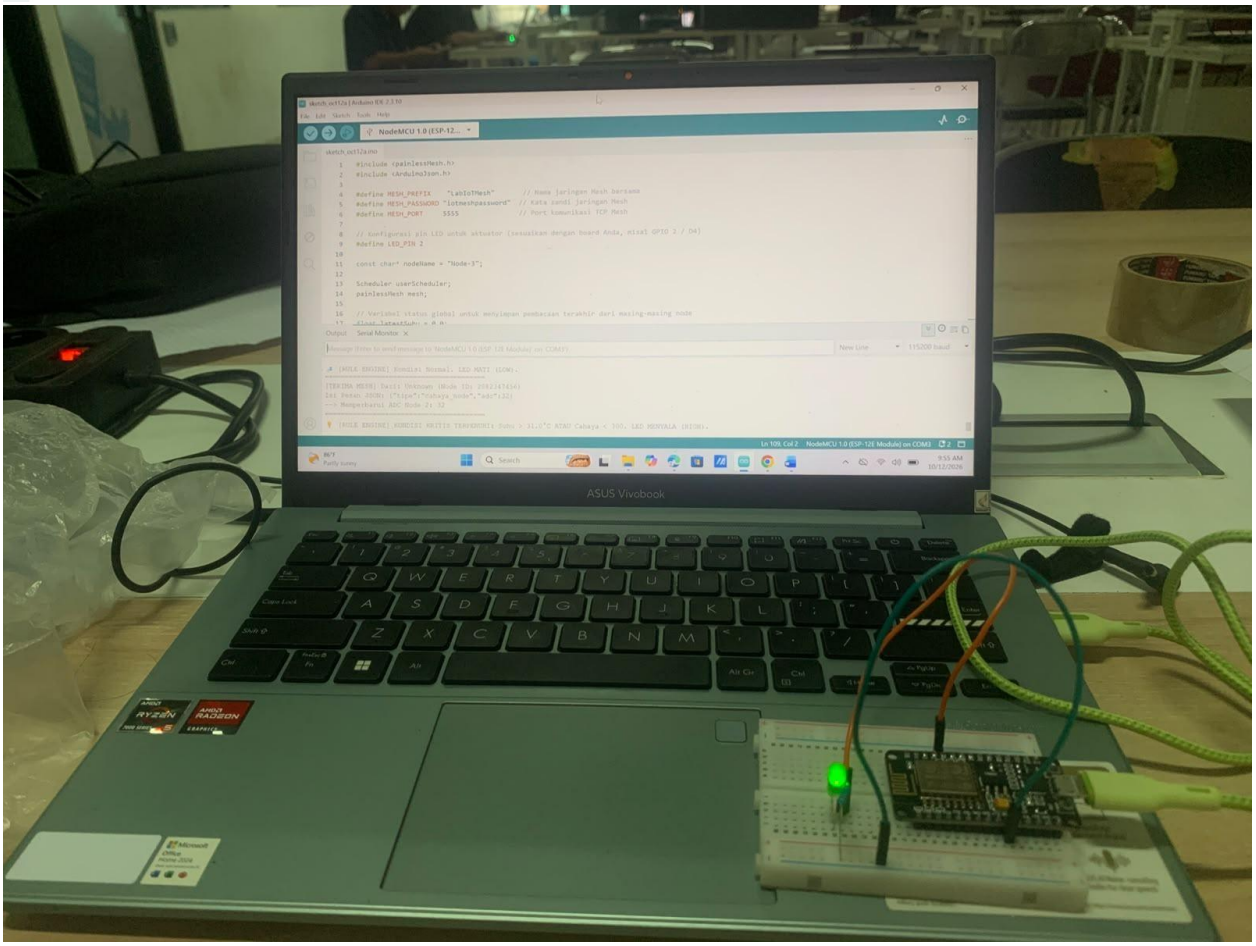
Scheduler userScheduler;
painlessMesh mesh;

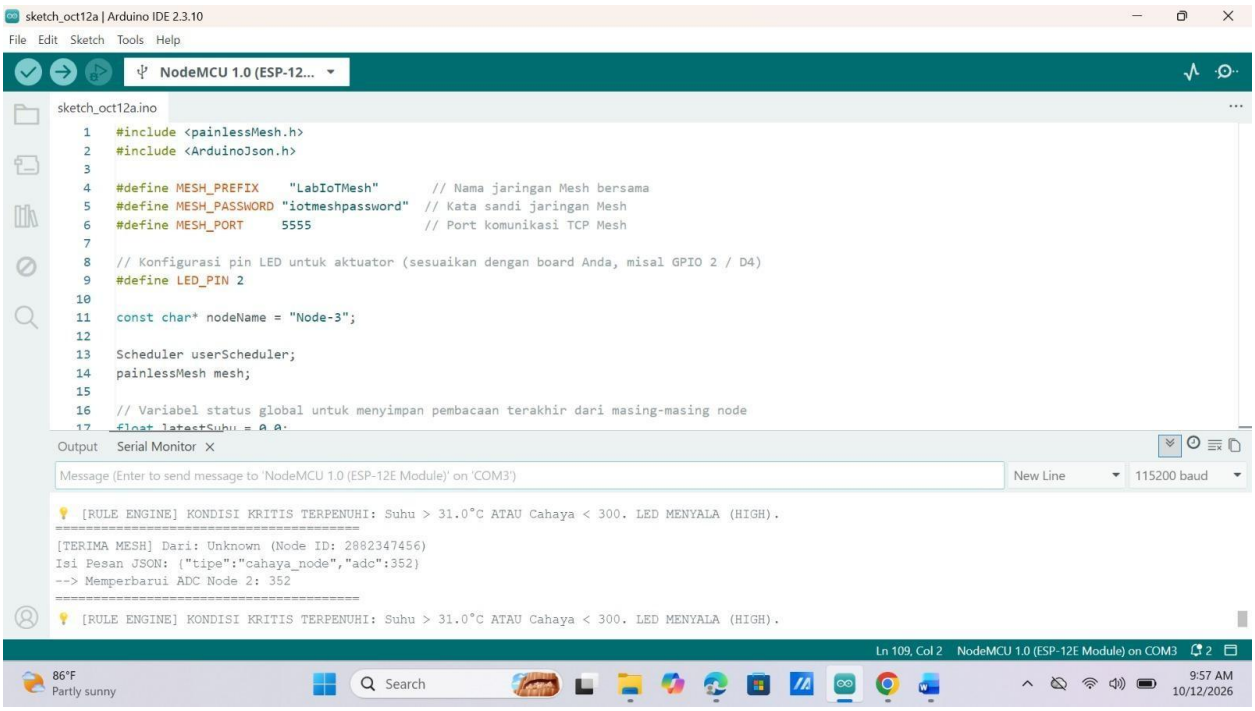
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
```

Output | Serial Monitor

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3') | New Line | 115200 baud

```
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.3,"kelembapan":40.2}
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.3,"kelembapan":39.9}
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.4,"kelembapan":39.3}
--> Koneksi Baru: 2743301850
--> Topologi mesh berubah
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.4,"kelembapan":38.9}
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.4,"kelembapan":39}
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.3,"kelembapan":38.8}
[KIRIM DHT] {"tipe":"suhu_node","suhu":29.4,"kelembapan":38.8}
```




sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "LabIoTMesh" // Nama jaringan Mesh bersama
#define MESH_PASSWORD "iotmeshpassword" // Kata sandi jaringan Mesh
#define MESH_PORT 5555 // Port komunikasi TCP Mesh
// Konfigurasi pin LED untuk aktuator (sesuaikan dengan board Anda, misal GPIO 2 / D4)
#define LED_PIN 2
const char* nodeName = "Node-3";
Scheduler userScheduler;
painlessMesh mesh;
// Variabel status global untuk menyimpan pembacaan terakhir dari masing-masing node
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
[RULE ENGINE] KONDISI KRITIS TERPENUHI: Suhu > 31.0°C ATAU Cahaya < 300. LED MENYALA (HIGH).
[TERIMA MESH] Dari: Unknown (Node ID: 2882347456)
Isi Pesan JSON: {"tipe":"cahaya_node","adc":352}
--> Memperbarui ADC Node 2: 352
[RULE ENGINE] KONDISI KRITIS TERPENUHI: Suhu > 31.0°C ATAU Cahaya < 300. LED MENYALA (HIGH).
Ln 109, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM3
86°F
Partly sunny
Search
9:57 AM
10/12/2026